

Lab 8

Name: Shiva Kumar

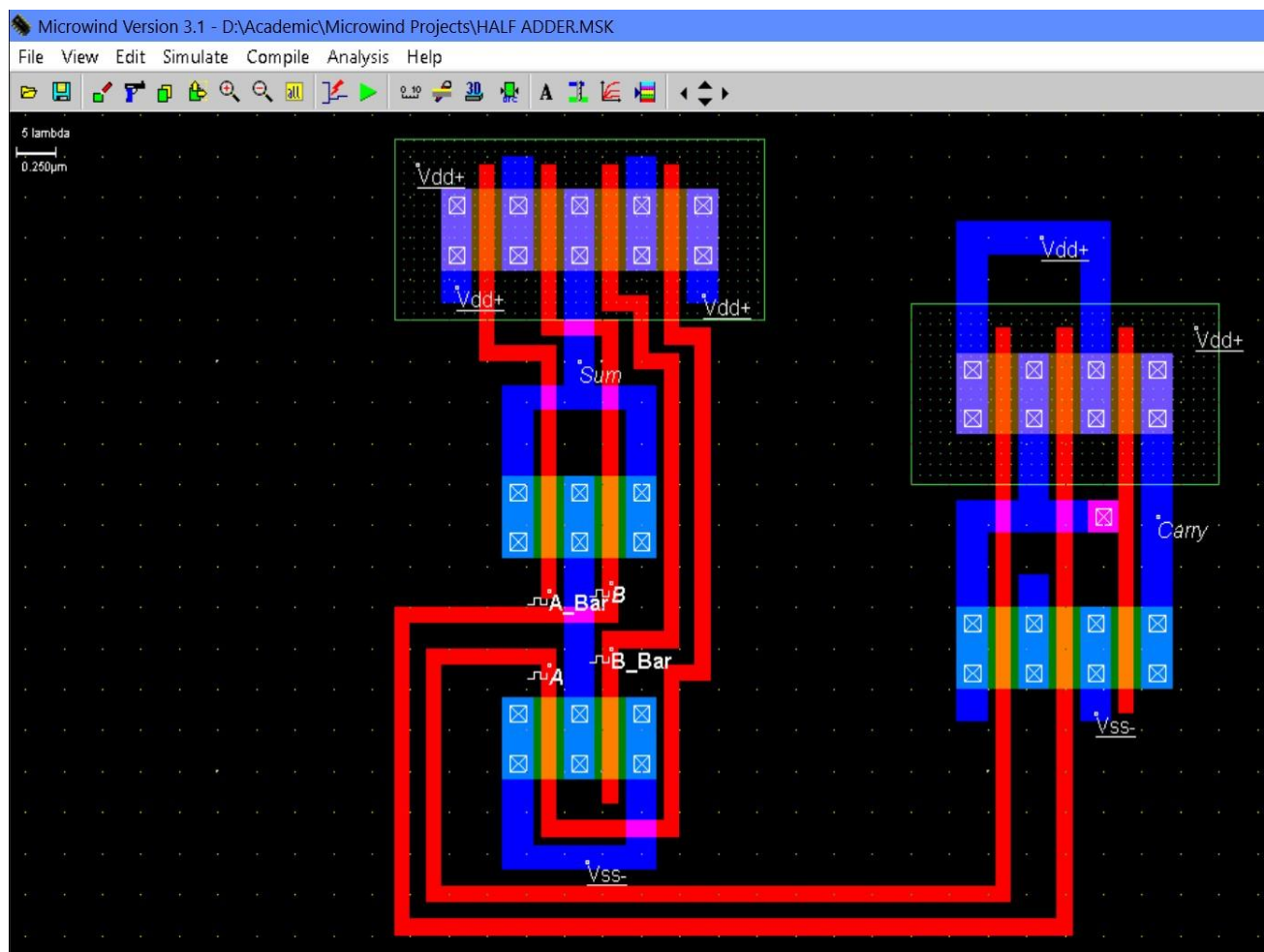
Enrollment No: 92301733056

CO1: Develop fundamentals in fabrication of MOSFETs and MOS transistor

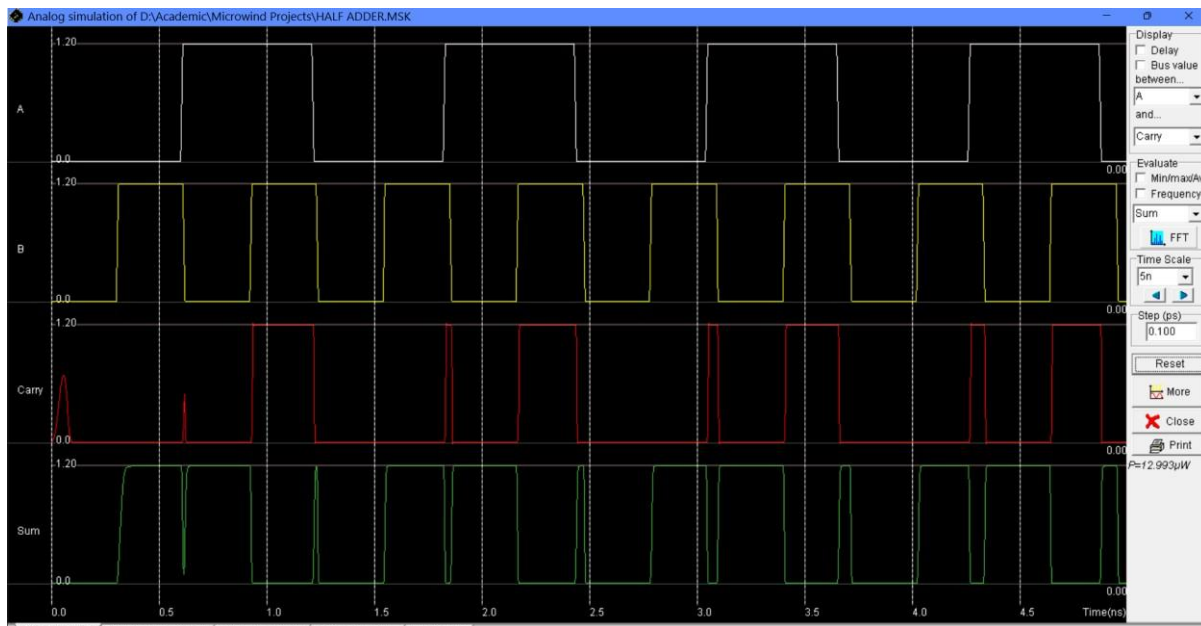
CO2: Analyze MOS inverters with its static and switching characteristics CO3: Analyze Combinational and Sequential MOS logic circuit

Experiment 8: To draw the layout of Half adder and Full adder simulate the design

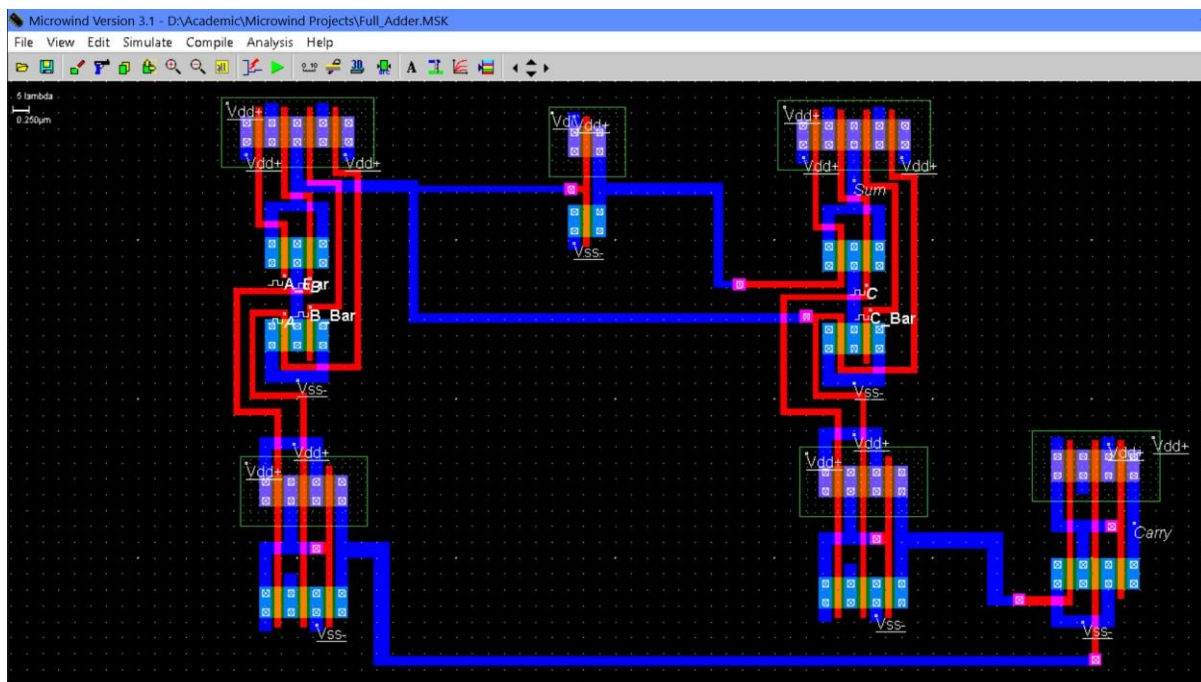
Half adder layout:



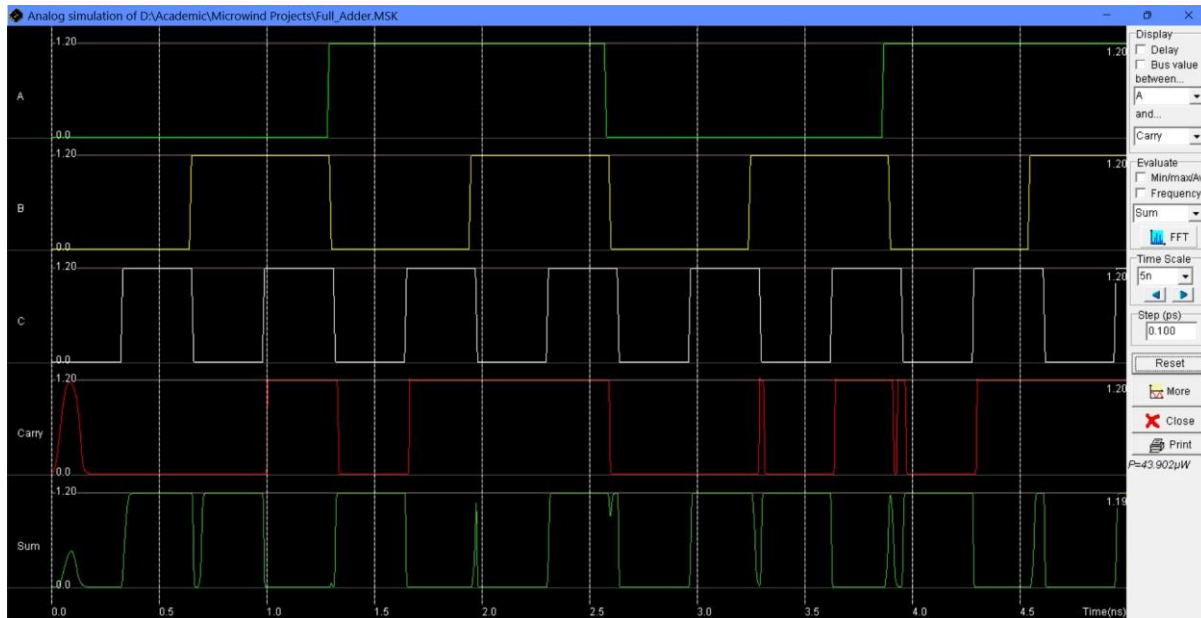
Waveforms:



Full adder layout:



Waveform:



Conclusion:- I have successfully designed the layout of a Half Adder and a Full Adder using Microwind. The default NMOS and PMOS layouts were used for transistor-level implementation. The Half Adder was realized using one XOR gate and one AND gate, while the Full Adder was implemented with two XOR gates, two AND gates, and one OR gate and NOT gate. I have verified the functionality of both adders by the waveforms, confirming that the outputs (Sum and Carry) matched with the expected truth table.